

Pediatric Medical Reference Guide

Comprehensive Guide to Common Childhood Conditions

This guide covers 14 common pediatric conditions including symptoms, causes, treatment, home care, when to see a doctor, and prevention.

Conditions: Fever, Ear Infections, Croup, RSV, Hand-Foot-and-Mouth Disease, Strep Throat, Pinkeye, Gastroenteritis, Common Cold, Asthma, Eczema, Head Lice, Chickenpox, Childhood Immunization.

Compiled from publicly available clinical guidelines: MedlinePlus (NIH/NLM), CDC, American Academy of Pediatrics, WHO, HealthyChildren.org.

■ **DISCLAIMER:** This document is for informational and educational purposes only. It is not a substitute for professional medical advice, diagnosis, or treatment.

1. Fever in Children

What Is a Fever

A fever is a body temperature that is higher than normal. A temperature above 100.4 degrees Fahrenheit (38 degrees Celsius) is generally considered a fever. Fever is not an illness itself but a sign that the body is fighting an infection. Fever is an important part of the body's defense against infection. Most fevers are caused by common viral infections and are not dangerous.

Normal body temperature varies throughout the day. It is typically lower in the morning and higher in the late afternoon. Normal ranges depend on measurement method: rectal temperatures read about 1 degree higher than oral, and oral reads about 1 degree higher than armpit (axillary) measurements.

Common Causes

Viral infections are the most common cause of fever in children, including colds, flu, and other respiratory viruses. Bacterial infections such as ear infections, urinary tract infections, strep throat, and pneumonia can also cause fever. Other causes include teething (may cause a very mild temperature rise), immunizations (low-grade fever for 1 to 2 days after some vaccines), overdressing an infant, and heat exposure.

Symptoms That May Accompany Fever

- Flushed or warm skin
- Sweating or chills and shivering
- Loss of appetite or refusal to eat
- Irritability, fussiness, or unusual drowsiness
- Decreased activity and general discomfort
- Dehydration signs: fewer wet diapers, no tears when crying, dry mouth

How to Take a Temperature

For infants under 3 months, a rectal temperature is most accurate. For children 3 months to 3 years, rectal or temporal artery (forehead) thermometers are recommended. For children over 4 years, oral or temporal artery thermometers work well. Ear (tympanic) thermometers can be used for children over 6 months. Armpit (axillary) temperatures are the least accurate but easiest to take.

Treatment and Home Care

A fever does not always need to be treated. If the child is active, playing, drinking fluids, and comfortable, treatment may not be necessary. Treatment is recommended when the child is uncomfortable.

- Keep the child hydrated with frequent small sips of clear fluids, breast milk, formula, or an oral rehydration solution like Pedialyte
- Dress the child in light, comfortable clothing. Do not bundle up a child with blankets or extra clothes even if shivering
- Keep the room at a comfortable temperature. A fan can help if the room is stuffy
- A lukewarm sponge bath can help reduce fever, but only when combined with fever-reducing medicine. Do not use cold baths, ice, or alcohol rubs

- Acetaminophen (Tylenol) can be given every 4 to 6 hours as needed. Follow package directions based on child's weight
- Ibuprofen (Advil, Motrin) can be given every 6 to 8 hours. Do not give ibuprofen to children younger than 6 months
- Never give aspirin to children or teenagers. Aspirin can cause Reye's syndrome, a rare but serious condition
- Do not alternate between acetaminophen and ibuprofen unless directed by a healthcare provider, as this can lead to dosing errors

Febrile Seizures

Some children experience seizures triggered by fever, called febrile seizures. These occur in about 2 to 5 percent of children between ages 6 months and 5 years. During a febrile seizure the child may shake, twitch, stiffen, or become unresponsive for a short time, usually less than 5 minutes. While frightening, most febrile seizures are harmless and do not cause lasting damage or epilepsy. If a seizure occurs: place the child on their side on a safe surface, do not put anything in their mouth, do not restrain them, time the seizure, and call 911 if it lasts more than 5 minutes.

When to Call the Doctor

■ Any infant under 3 months with a rectal temperature of 100.4 F (38 C) or higher needs immediate medical evaluation, even if no other symptoms are present

- Child 3 to 12 months with a fever of 102.2 F (39 C) or higher
- Child under 2 years with a fever lasting more than 48 hours
- Fever over 104 F (40 C) at any age
- Fever that keeps coming back over a week or more
- Child appears very ill, is unusually sleepy, or is very fussy
- Signs of dehydration: no wet diapers for 8 hours, no tears when crying, dry mouth, sunken eyes
- New rash accompanying the fever
- Stiff neck, severe headache, difficulty breathing, blue lips
- Child has a chronic medical condition or weakened immune system

Prevention

Frequent handwashing is the most effective way to prevent infections that cause fever. Keep children up to date on immunizations. Teach children to cover coughs and sneezes. Avoid sharing cups and utensils. Keep children home when they are sick to prevent spreading illness.

Source: MedlinePlus/NIH, CDC Fever Guidelines, AAP HealthyChildren.org

2. Ear Infections (Otitis Media)

What Is an Ear Infection

An ear infection (acute otitis media) is an infection of the middle ear, the air-filled space behind the eardrum. Ear infections are the most common reason parents bring their child to a doctor. Five out of six children will have at least one ear infection by their third birthday. They are most common in children aged 6 months to 2 years. The Eustachian tube connects the middle ear to the back of the throat and normally drains fluid. In young children, these tubes are shorter, more horizontal, and narrower, making them easier to block and harder to drain. When the tube becomes blocked, fluid builds up and bacteria or viruses can grow, causing infection.

Types of Ear Infections

- Acute otitis media (AOM): the most common type, where the middle ear is infected and swollen with trapped fluid behind the eardrum, causing pain and often fever
- Otitis media with effusion (OME): fluid remains behind the eardrum after an infection clears, or builds up without active infection. Usually not painful but can affect hearing
- Chronic otitis media with effusion: fluid remains for an extended period or returns repeatedly without infection, which can affect hearing and language development
- Swimmer's ear (otitis externa): an infection of the outer ear canal, different from a middle ear infection

Symptoms

- Ear pain (tugging or pulling at the ear in infants)
- Fussiness, irritability, or difficulty sleeping
- Fever, especially in younger children
- Fluid draining from the ear (yellow or white discharge)
- Difficulty hearing or responding to sounds
- Loss of balance or clumsiness
- Loss of appetite (swallowing can cause pressure changes that are painful)

Ear infections often follow or accompany an upper respiratory infection such as a cold.

Treatment

Many mild ear infections clear on their own without antibiotics. The AAP and CDC recommend a wait-and-see approach for some children, giving the immune system 48 to 72 hours to fight the infection before starting antibiotics.

- Watchful waiting is appropriate for children 6 months and older with mild symptoms and a temperature below 102.2 F
- Antibiotics are typically prescribed immediately for: all children under 6 months, children with severe symptoms (high fever, severe pain, both ears infected), and children whose symptoms do not improve within 48 to 72 hours
- Amoxicillin is the most commonly prescribed antibiotic for ear infections. If prescribed, complete the full course even if symptoms improve
- Pain management: acetaminophen or ibuprofen for pain and fever. Warm compresses held against the ear can provide comfort. Some providers may prescribe numbing ear drops

- If antibiotics do not improve symptoms within 48 to 72 hours, contact the provider as a different antibiotic may be needed

Home Care

- Apply a warm, moist washcloth over the affected ear for 10 to 15 minutes to ease pain
- Keep the child's head elevated during rest
- Encourage swallowing by offering fluids, which can help the Eustachian tube drain
- Do not insert anything into the ear canal
- If prescribed ear drops, warm the bottle in your hands first for comfort

Ear Tubes

Children who have frequent ear infections (three or more in six months, or four or more in a year) or persistent fluid that affects hearing may benefit from ear tube surgery (myringotomy). A surgeon places tiny tubes through the eardrums to allow air flow and fluid drainage. The tubes usually stay in place for 6 to 18 months and fall out on their own. The procedure is one of the most common childhood surgeries and typically takes about 15 minutes.

When to See a Doctor

- Any infant under 6 months with suspected ear infection
- Ear pain lasting more than 2 days
- Fever above 102.2 F with ear symptoms
- Pus or bloody discharge from the ear
- Symptoms that worsen or do not improve within 48 to 72 hours
- Hearing concerns after infection resolves
- Swelling, redness, or pain behind the ear (possible mastoiditis, a serious complication)

Prevention

- Keep immunizations current, including pneumococcal and flu vaccines
- Breastfeed for at least 6 months if possible, as breast milk contains antibodies
- Do not prop bottles or allow infants to drink lying flat
- Keep children away from secondhand smoke
- Practice good hand hygiene and reduce exposure to sick children

Source: MedlinePlus/NIH, NIDCD, CDC Ear Infection Guidelines, AAP Otitis Media Guidelines

3. Croup

What Is Croup

Croup is a common childhood illness that causes swelling of the voice box (larynx), windpipe (trachea), and bronchial tubes (bronchi). It causes a distinctive barking cough that sounds like a seal. Croup is most common in children aged 6 months to 3 years, with peak incidence around age 2. It most often occurs in fall and winter. Croup is usually caused by a viral infection, most commonly parainfluenza virus. Other viruses including RSV, influenza, and adenovirus can also cause croup.

Symptoms

- A harsh, barking cough (the hallmark symptom, often described as sounding like a seal)
- Stridor: a high-pitched, squeaky, or whistling sound when breathing in (inspiratory stridor)
- Hoarse or raspy voice
- Fever, usually low-grade but can be higher
- Runny nose and cold-like symptoms (often precede the cough by 1 to 2 days)
- Symptoms typically worsen at night and with crying or agitation
- Symptoms usually peak on the second or third night and resolve within a week

Severity Levels

Mild croup: occasional barking cough, no stridor at rest, no significant breathing difficulty. Moderate croup: frequent barking cough, stridor at rest, mild chest retractions (skin pulling in between ribs or at the neck).

Severe croup: significant stridor at rest, marked retractions, agitation, decreased air entry, possible cyanosis (blue coloring). Severe croup requires emergency care.

Treatment and Home Care

- Keep the child calm and comfortable. Crying and agitation worsen symptoms significantly
- Cool mist: take the child outside into cool night air for 10 to 15 minutes, or sit in a bathroom with cool mist from the shower. Cool moist air can help reduce airway swelling
- Steam: sit in a closed bathroom with a hot shower running to create steam. This can help some children, though evidence is mixed
- Keep the child upright or propped up to ease breathing
- Offer warm clear fluids to soothe the throat and keep the child hydrated
- Acetaminophen or ibuprofen for fever and discomfort
- For moderate to severe croup, a doctor may prescribe oral dexamethasone (a steroid) to reduce airway swelling. A single dose often provides dramatic improvement
- In emergency settings, nebulized epinephrine may be given for severe symptoms. This provides rapid but temporary relief

When to Go to the ER

■ Call 911 immediately if the child has blue or gray lips or fingertips, is struggling to breathe, cannot speak or cry due to breathing difficulty, or is drooling and unable to swallow

- Stridor that is getting louder or does not stop at rest

- Visible retractions: the skin between the ribs, above the collarbone, or at the neck is pulling in with each breath
- The child appears anxious, very restless, or unusually still and tired
- Difficulty swallowing or drooling (could indicate epiglottitis, a medical emergency)
- Fever above 104 F or the child looks very ill

Prevention

Croup is caused by contagious viruses spread through coughing, sneezing, and respiratory droplets. Frequent handwashing, avoiding close contact with sick individuals, and keeping children home when ill can reduce spread. There is no vaccine specifically for croup, but the flu vaccine can prevent flu-related croup.

Source: MedlinePlus/NIH, AAP Croup Guidelines, HealthyChildren.org

4. RSV (Respiratory Syncytial Virus)

What Is RSV

Respiratory syncytial virus (RSV) is a very common virus that infects the respiratory tract. Nearly all children are infected with RSV at least once by age 2. In most older children and adults, RSV causes mild cold-like symptoms. However, RSV can be serious in infants (especially those born premature), young children under 2 years, and children with chronic lung or heart disease or weakened immune systems. RSV is the most common cause of bronchiolitis (inflammation of the small airways in the lungs) and pneumonia in children under 1 year of age in the United States. RSV season typically runs from fall through spring.

Symptoms

Symptoms usually appear 4 to 6 days after exposure and develop in stages.

- Early symptoms (1 to 3 days): runny nose, decreased appetite, cough that may gradually worsen
- Progressing symptoms: sneezing, fever, wheezing
- In young infants: irritability, decreased activity, poor feeding, and apnea (pauses in breathing)
- Bronchiolitis signs: rapid breathing, flaring of the nostrils, head bobbing with breathing, rhythmic grunting during breathing, belly breathing and tugging between the ribs or lower neck, wheezing

Treatment and Home Care

There is no specific antiviral treatment for RSV. Treatment focuses on managing symptoms and keeping the child comfortable and hydrated.

- Keep the child well hydrated with frequent, small feedings of breast milk, formula, or Pedialyte. Infants may feed more slowly and need smaller, more frequent feeds
- Use a bulb syringe or nasal aspirator to gently suction mucus from the nose, especially before feeding and sleeping
- Use saline nose drops to loosen thick nasal mucus before suctioning
- Run a cool-mist humidifier in the room to keep air moist and help with congestion
- Acetaminophen or ibuprofen (age 6 months and older) for fever and discomfort. Never give aspirin
- Keep the child upright or with head slightly elevated to ease breathing
- Do not give over-the-counter cold medicines to children under 4 years

When to Seek Emergency Care

■ RSV can progress rapidly in infants. Seek emergency care immediately for any signs of breathing difficulty

- Rapid breathing: more than 60 breaths per minute in infants
- Labored breathing: nostrils flaring, ribs showing with each breath, belly moving exaggeratedly
- Wheezing or grunting sounds with each breath
- Blue or gray color of lips, mouth, or fingernails (sign of low oxygen)
- Pauses in breathing lasting more than 10 seconds
- High fever in an infant under 3 months
- Signs of dehydration: fewer than 1 wet diaper every 8 hours, no tears when crying, unusually sleepy
- The child is unable or unwilling to drink fluids

Prevention

- Frequent handwashing with soap and water for at least 20 seconds
- Avoid touching the face with unwashed hands
- Avoid close contact with sick people, especially kissing on the face
- Clean and disinfect frequently touched surfaces
- Keep infants away from crowds and daycare settings during RSV season if possible
- Do not expose children to cigarette smoke
- Nirsevimab (Beyfortus) is an immunization recommended for infants entering their first RSV season and some high-risk toddlers
- Pregnant persons can receive an RSV vaccine (Abrysvo) between 32 and 36 weeks of pregnancy to help protect newborns

Source: MedlinePlus/NIH, CDC RSV Guidelines, AAP RSV Prevention Policy

5. Hand, Foot, and Mouth Disease

What Is Hand, Foot, and Mouth Disease

Hand, foot, and mouth disease (HFMD) is a common viral illness in infants and children under 5 years, though older children and adults can get it too. It is caused by viruses in the enterovirus group, most commonly coxsackievirus A16 and enterovirus 71. HFMD is not the same as foot-and-mouth disease in animals. It spreads through close personal contact, respiratory droplets, contact with fluid from blisters, and contact with contaminated surfaces or objects. It is most contagious during the first week of illness. Outbreaks are most common in summer and early fall.

Symptoms

Symptoms usually appear 3 to 6 days after exposure.

- Fever, often the first symptom, usually 101 to 103 F
- Sore throat and decreased appetite
- Feeling of being unwell (malaise)
- Painful sores in the mouth: start as red spots on the tongue, gums, and inside of the cheeks that develop into small blisters and then painful ulcers. These make eating and drinking difficult
- Skin rash: flat or raised red spots, sometimes with blisters, on the palms of the hands, soles of the feet, and sometimes the buttocks, knees, elbows, and genital area. The rash is not usually itchy
- In some cases, fingernail and toenail loss can occur weeks after the illness (temporary and nails regrow)

Treatment and Home Care

There is no specific treatment or antiviral medication for HFMD. The illness typically resolves on its own in 7 to 10 days. Treatment focuses on comfort and preventing dehydration.

- Pain and fever relief: acetaminophen or ibuprofen (not aspirin)
- Cold foods: popsicles, ice cream, cold yogurt, and smoothies soothe mouth sores and encourage fluid intake
- Avoid acidic, salty, or spicy foods and drinks (orange juice, tomato sauce, salsa) as they can irritate mouth sores
- Offer frequent sips of cold water, milk, or Pedialyte to prevent dehydration
- For older children, a mixture of equal parts liquid antacid (like Maalox) and liquid diphenhydramine (Benadryl) can be swished and spit to numb mouth pain. Ask the pediatrician about this first
- Keep blisters clean and uncovered. They will heal on their own
- Soft foods are easier to eat: applesauce, mashed potatoes, soup (lukewarm, not hot)

When to See a Doctor

- Child refuses to drink or shows signs of dehydration (no tears, dry mouth, no urination for 8 hours)
- Fever lasts more than 3 days
- Symptoms worsen instead of improving after 7 to 10 days
- Mouth sores are severe enough to prevent any fluid intake
- Child is very young (under 6 months)

- Child has a weakened immune system

Contagion and Prevention

Children are most contagious during the first week but can continue to spread the virus for weeks through stool. The virus can also live on surfaces for hours.

- Frequent handwashing with soap and water, especially after diaper changes and using the toilet
- Disinfect frequently touched surfaces and shared toys
- Avoid close contact such as kissing, hugging, and sharing cups and utensils with infected persons
- Keep infected children home from daycare or school until fever is gone and mouth sores have healed

Source: MedlinePlus/NIH, CDC HFMD Information, AAP HealthyChildren.org

6. Strep Throat

What Is Strep Throat

Strep throat is a bacterial infection of the throat and tonsils caused by Group A Streptococcus (GAS) bacteria. It is most common in children ages 5 to 15 but can occur at any age. Strep throat is uncommon in children under 3 years. It spreads through respiratory droplets when an infected person coughs or sneezes, and through shared food or drinks. Strep throat is different from a viral sore throat, which is more common and does not require antibiotics.

Symptoms

Strep throat symptoms usually appear 2 to 5 days after exposure and come on suddenly.

- Severe sore throat that comes on quickly (not gradual like a cold)
- Pain and difficulty swallowing
- Fever, often 101 F or higher
- Red, swollen tonsils, sometimes with white patches or streaks of pus
- Tiny red spots (petechiae) on the roof of the mouth
- Swollen, tender lymph nodes in the front of the neck
- Headache, stomachache, nausea, or vomiting (especially in younger children)
- Body aches and general feeling of being unwell
- Sometimes a fine, red, sandpaper-like rash (scarlet fever)

Symptoms that suggest a viral sore throat rather than strep include: cough, runny nose, hoarseness, conjunctivitis (pink eye), and diarrhea. Strep throat typically does NOT cause cough or runny nose.

Diagnosis

Strep throat cannot be diagnosed by looking at the throat alone. A rapid strep test or throat culture is needed. The rapid test takes about 5 to 10 minutes. If the rapid test is negative but the doctor still suspects strep, a throat culture may be sent to a lab (results in 1 to 2 days). Testing is important because strep requires antibiotic treatment to prevent complications.

Treatment

- Antibiotics are essential for strep throat. Penicillin or amoxicillin are the most commonly prescribed. For penicillin-allergic patients, alternatives include azithromycin, cephalexin, or clindamycin
- The full course of antibiotics (usually 10 days) must be completed even if symptoms improve, to fully eliminate the bacteria and prevent complications
- Children are no longer contagious after 12 to 24 hours of antibiotic treatment and can return to school or daycare after at least 12 hours on antibiotics and no fever
- Pain and fever management: acetaminophen or ibuprofen
- Warm liquids like broth, warm water with honey (children over 1 year), or warm tea can soothe the throat
- Cold foods like popsicles and ice chips can also help with pain
- Gargling with warm salt water (older children who can gargle without swallowing)
- Rest and adequate hydration

Complications of Untreated Strep

Untreated strep throat can lead to serious complications. This is why antibiotic treatment is important.

- Rheumatic fever: an inflammatory disease that can affect the heart, joints, nervous system, and skin. Can cause permanent heart damage
- Post-streptococcal glomerulonephritis: kidney inflammation
- Peritonsillar abscess: a pocket of pus beside the tonsil
- Spread of infection to the blood, skin, or middle ear

When to See a Doctor

- Sore throat with fever, especially without cold symptoms like cough or runny nose
- Sore throat with swollen lymph nodes
- Sore throat that lasts more than 48 hours
- Difficulty swallowing or breathing
- Rash accompanying a sore throat
- Known exposure to someone with strep throat

Prevention

- Wash hands frequently with soap and water
- Do not share eating utensils, cups, or water bottles
- Teach children to cover mouth and nose when coughing or sneezing
- Replace toothbrush after starting antibiotics to prevent reinfection

Source: MedlinePlus/NIH, CDC Group A Strep, AAP Strep Throat Guidelines

7. Pinkeye (Conjunctivitis)

What Is Conjunctivitis

Conjunctivitis, commonly called pinkeye, is an inflammation or infection of the conjunctiva, the clear membrane that lines the inside of the eyelid and covers the white part of the eye. When the small blood vessels in the conjunctiva become inflamed, they become more visible, giving the eye a pink or reddish appearance. Pinkeye is very common in children and can be caused by viruses, bacteria, allergens, or irritants.

Types and Their Symptoms

Viral conjunctivitis: the most common type, often associated with a cold or upper respiratory infection. Affects one or both eyes. Produces watery discharge. Eyes are red and may be itchy. Highly contagious, spreads through contact with eye discharge. Usually resolves on its own in 7 to 14 days.

Bacterial conjunctivitis: produces thick, yellow-green discharge that may cause the eyelids to stick together, especially upon waking. Can affect one or both eyes. Eyes are red and may be swollen. Contagious through direct contact with discharge. May require antibiotic eye drops or ointment.

Allergic conjunctivitis: affects both eyes. Intense itching, tearing, and swelling. Often accompanied by sneezing, runny nose, and other allergy symptoms. Not contagious. Triggered by pollen, dust mites, pet dander, or other allergens.

Treatment and Home Care

- **Viral:** no specific treatment. Cool compresses and artificial tears (lubricating eye drops) for comfort. Resolves on its own in 1 to 2 weeks
- **Bacterial:** antibiotic eye drops or ointment prescribed by a doctor. Apply as directed for the full course, usually 5 to 7 days
- **Allergic:** avoid the allergen if possible. Over-the-counter antihistamine eye drops or oral antihistamines can help. Cool compresses relieve itching
- **For all types:** gently clean discharge from the eye using a warm, damp cotton ball or clean washcloth. Wipe from the inner corner outward. Use a fresh cotton ball for each wipe and for each eye
- Do not touch or rub the eyes
- Wash hands thoroughly after touching the eyes or face
- Do not share towels, pillowcases, or eye makeup
- Do not wear contact lenses until the infection clears and for 24 hours after finishing antibiotic drops

When to See a Doctor

- Moderate to severe eye pain (beyond mild irritation)
- Sensitivity to light or blurred vision that does not clear with blinking
- Intense redness in one or both eyes
- Symptoms that do not improve within 3 to 4 days or worsen
- Pinkeye in a newborn (needs immediate evaluation to rule out serious bacterial infection)
- Child has a weakened immune system

Contagion and School

Viral and bacterial pinkeye are contagious. Children with bacterial conjunctivitis can usually return to school 24 hours after starting antibiotic drops. Many schools allow children with viral conjunctivitis to attend as long as they can avoid touching their eyes and practice good hand hygiene. Check with your school's policy.

Source: MedlinePlus/NIH, CDC Conjunctivitis Information, AAP HealthyChildren.org

8. Gastroenteritis (Stomach Flu)

What Is Gastroenteritis

Gastroenteritis is inflammation of the stomach and intestines, commonly called the stomach flu or stomach bug, though it is not related to influenza. It is very common in children. The most common cause in children is viral infection, particularly rotavirus and norovirus. Bacterial causes include Salmonella, E. coli, and Campylobacter. It spreads through contaminated food or water, contact with an infected person, or touching contaminated surfaces. The biggest danger of gastroenteritis in children is dehydration from fluid loss through vomiting and diarrhea.

Symptoms

- Watery diarrhea (most common symptom)
- Vomiting, often before diarrhea begins
- Stomach cramps and abdominal pain
- Nausea and loss of appetite
- Low-grade fever
- Muscle aches and headache
- Symptoms typically last 1 to 3 days for viral gastroenteritis but can last up to 10 days

Treatment and Home Care

The primary goal is to prevent dehydration by replacing lost fluids and electrolytes.

- Oral rehydration solution (Pedialyte, Enfalyte) is the best choice for rehydration. It contains the right balance of water, sugar, and salts
- Start with small, frequent sips: 1 teaspoon (5 mL) every 1 to 2 minutes. Gradually increase as tolerated
- For breastfed infants: continue breastfeeding, offer more frequently
- For formula-fed infants: offer oral rehydration solution and continue formula as tolerated
- Avoid fruit juice, soda, sports drinks, and undiluted broth, as the sugar content can worsen diarrhea
- Once vomiting subsides (usually 6 to 12 hours), gradually reintroduce regular foods. The BRAT diet (bananas, rice, applesauce, toast) is commonly suggested, but any bland, easy-to-digest foods are fine
- Avoid dairy products, fatty foods, and sugary foods initially
- Acetaminophen for fever and discomfort. Avoid ibuprofen if the child is dehydrated or vomiting
- Do not give anti-diarrheal medications (such as Imodium) to young children unless directed by a doctor

Signs of Dehydration (Requires Medical Attention)

■ Dehydration is the main danger of gastroenteritis, especially in infants and young children

- No wet diapers for 6 or more hours (infants) or no urination for 8 hours (older children)
- No tears when crying
- Dry mouth and tongue
- Sunken eyes or sunken soft spot (fontanelle) on an infant's head
- Unusual drowsiness, lethargy, or decreased alertness

- Skin that does not bounce back when gently pinched

When to See a Doctor

- Any infant under 3 months with vomiting or diarrhea
- Signs of dehydration listed above
- Vomiting that lasts more than 24 hours
- Diarrhea that lasts more than a few days or contains blood or mucus
- Fever above 102 F that does not respond to fever-reducing medication
- Severe abdominal pain
- The child appears very ill or cannot keep any fluids down
- Child has a chronic health condition

Prevention

- Rotavirus vaccine: given to infants starting at 2 months of age, protects against the most common cause of severe gastroenteritis in young children
- Wash hands thoroughly with soap and water, especially after diaper changes, using the toilet, and before eating or preparing food
- Disinfect contaminated surfaces with a bleach-based cleaner
- Keep sick children home from daycare or school until 24 hours after the last episode of vomiting or diarrhea
- Wash soiled clothing and bedding promptly in hot water

Source: MedlinePlus/NIH, CDC Viral Gastroenteritis, AAP Gastroenteritis Management

9. Common Cold

What Is the Common Cold

The common cold is a viral infection of the upper respiratory tract (nose and throat). Over 200 different viruses can cause colds, with rhinoviruses being the most common. Children average 6 to 8 colds per year, and children in daycare may have even more. Colds are more frequent in fall and winter. Most colds resolve on their own in 7 to 10 days, though some symptoms may linger for up to 2 weeks. Colds are spread through airborne droplets, direct contact with an infected person, or touching contaminated surfaces and then touching the nose, mouth, or eyes.

Symptoms

Cold symptoms usually develop 1 to 3 days after exposure to the virus.

- Runny or stuffy nose (discharge may start clear and become thicker and yellow or green as the cold progresses; this color change is normal and does not necessarily mean a bacterial infection)
- Sneezing
- Cough (may worsen at night)
- Sore throat (usually mild)
- Mild body aches
- Low-grade fever (more common in younger children)
- Decreased appetite
- Watery eyes
- General fussiness and irritability in infants

Treatment and Home Care

There is no cure for the common cold. Antibiotics do not work against viruses and should not be given for colds. Treatment focuses on relieving symptoms.

- Rest and fluids: encourage the child to drink plenty of water, diluted juice, breast milk, formula, or warm broth
- Nasal saline drops or spray to loosen nasal mucus, followed by gentle suctioning with a bulb syringe (especially helpful for infants before feeding and sleeping)
- Cool-mist humidifier in the bedroom to add moisture to the air and ease congestion
- Honey for cough: one-half to one teaspoon of honey can soothe a cough in children over 12 months. Never give honey to infants under 1 year due to risk of botulism
- Acetaminophen or ibuprofen for fever and discomfort. Do not give aspirin
- Do not give over-the-counter cough and cold medicines to children under 4 years old. They are not proven effective and can have serious side effects
- For older children: throat lozenges, gargling with warm salt water, and nasal decongestant sprays (for short-term use only, 3 days max) may provide some relief
- Elevate the head of the mattress slightly to help with nighttime congestion

When to See a Doctor

- Any cold symptoms in an infant under 3 months
- Fever above 100.4 F in an infant under 3 months
- Fever lasting more than 3 days in any age
- Symptoms lasting more than 10 days without improvement
- Symptoms that worsen after initial improvement (could suggest secondary bacterial infection)
- Ear pain or drainage from the ear
- Difficulty breathing or rapid breathing
- Persistent cough lasting more than 3 weeks
- Child is excessively sleepy or unusually fussy

Prevention

- Frequent handwashing with soap and water for at least 20 seconds
- Teach children to cough or sneeze into their elbow, not their hands
- Avoid close contact with people who are sick
- Regularly clean and disinfect frequently touched surfaces and toys
- Do not share cups, utensils, or towels
- There is no vaccine for the common cold

Source: MedlinePlus/NIH, CDC Common Cold Information, AAP HealthyChildren.org

10. Asthma in Children

What Is Asthma

Asthma is a chronic lung disease that inflames and narrows the airways (bronchial tubes). It is the most common chronic disease in children, affecting about 6 million children in the United States. In asthma, the airways are always somewhat inflamed, and during an asthma attack (exacerbation or flare-up), the airway walls swell further, the muscles around the airways tighten (bronchospasm), and extra mucus is produced, all of which make breathing difficult. Asthma cannot be cured, but it can be managed effectively with proper treatment. Many children with asthma can live active, normal lives.

Symptoms

- Wheezing (a whistling sound when breathing, especially when breathing out)
- Coughing that worsens at night, during exercise, or with cold air, laughing, or crying
- Chest tightness or pain (younger children may describe this as a tummy ache)
- Shortness of breath or rapid breathing
- Difficulty sleeping due to coughing or breathing difficulty
- Fatigue during play or exercise (child stops to rest more than peers)
- In infants: poor feeding, grunting during feeding, decreased activity

Common Triggers

- Respiratory infections (colds, flu, RSV)
- Allergens: pollen, dust mites, mold, pet dander, cockroach droppings
- Irritants: tobacco smoke, air pollution, strong odors, chemical fumes
- Exercise or physical activity (exercise-induced bronchoconstriction)
- Cold air and weather changes
- Strong emotions (laughing hard, crying, stress)
- Gastroesophageal reflux (GERD)

Treatment

Asthma treatment involves two main types of medications and avoiding triggers.

- Quick-relief (rescue) medications: bronchodilators like albuterol (ProAir, Ventolin) relax airway muscles and provide rapid relief during an asthma attack. Used with an inhaler and spacer. Should be taken at the first sign of symptoms
- Long-term controller medications: inhaled corticosteroids (fluticasone, budesonide) reduce airway inflammation and prevent flare-ups. Taken daily even when feeling well
- Asthma action plan: a written plan from the doctor that tells you which medications to take, when to take them, when to increase doses, and when to seek emergency care. Uses a green-yellow-red zone system
- Spacer device: a chamber that attaches to the inhaler to help deliver medication more effectively to the lungs. Recommended for all children using metered-dose inhalers
- Nebulizer: a machine that turns liquid medicine into a mist for infants and young children who cannot use an inhaler effectively

During an Asthma Attack

- Give the quick-relief (rescue) inhaler immediately. Use 2 to 4 puffs with a spacer
- Wait 20 minutes. If symptoms do not improve, give another 2 to 4 puffs
- Keep the child calm and in an upright sitting position
- Follow the asthma action plan

When to Go to the ER

■ Call 911 if the child is gasping for air, cannot speak in full sentences, has blue lips or fingernails, or the rescue inhaler is not helping

- Severe difficulty breathing or child is hunched over and struggling
- Rescue inhaler provides no relief after two treatments
- Nostrils are flaring wide with each breath
- Ribs or neck muscles are visible with each breath (retractions)

Prevention and Management

- Identify and avoid triggers specific to the child
- Take controller medications daily as prescribed, even when symptom-free
- Get a flu vaccine every year
- Keep the home smoke-free
- Use allergy-proof mattress and pillow covers if dust mites are a trigger
- Monitor symptoms and peak flow measurements regularly
- Regular follow-up with the doctor to adjust the treatment plan

Source: MedlinePlus/NIH, CDC Asthma in Children, AAP Asthma Guidelines

11. Eczema (Atopic Dermatitis)

What Is Eczema

Eczema, also called atopic dermatitis, is a chronic skin condition that causes dry, itchy, inflamed skin. It is very common in children, affecting up to 15 to 20 percent of children worldwide. Eczema often begins in infancy or early childhood (usually before age 5) and may improve or resolve with age, though some people have it into adulthood. Eczema is not contagious. It is related to an overactive immune response and a weakened skin barrier that allows moisture to escape and irritants and allergens to enter. Children with eczema have a higher risk of developing food allergies, hay fever, and asthma (called the atopic march).

Symptoms

- Dry, sensitive skin
- Intense itching, which can worsen at night and disrupt sleep
- Red to brownish-gray patches, especially on the cheeks, arms, and legs in infants and in the creases of elbows, knees, and neck in older children
- Small, raised bumps that may leak fluid when scratched and then crust over
- Thickened, cracked, or scaly skin from chronic scratching
- Raw, swollen, sensitive skin from scratching

Flare-ups (periods when symptoms worsen) alternate with periods of remission (when skin clears or improves).

Common Triggers

- Dry skin and low humidity
- Irritants: soaps, detergents, shampoos, disinfectants, fragrances
- Rough or scratchy fabrics (wool, synthetic materials)
- Allergens: dust mites, pollen, mold, pet dander
- Food allergens (in some children): eggs, milk, peanuts, soy, wheat
- Heat, sweating, and sudden temperature changes
- Stress and emotional factors
- Skin infections (bacterial, viral, or fungal)

Treatment and Home Care

The cornerstone of eczema management is moisturizing the skin and avoiding triggers.

- Moisturize frequently and generously: apply a thick, fragrance-free cream or ointment (such as petroleum jelly, CeraVe, Eucerin, or Aquaphor) at least twice daily and immediately after bathing. Ointments and creams work better than lotions
- Bathing routine: lukewarm (not hot) baths lasting 5 to 10 minutes. Use a gentle, fragrance-free, soap-free cleanser only on dirty areas. Pat skin dry gently (do not rub) and apply moisturizer within 3 minutes while skin is still damp
- Topical corticosteroids: low-strength over-the-counter hydrocortisone cream (1 percent) can be used on mild flares for short periods. Stronger prescription corticosteroids may be needed for moderate to severe flares. Apply a thin layer only to affected areas as directed

- Avoid scratching: keep fingernails short, use cotton gloves at night for young children, distract the child during itchy episodes
- Dress the child in soft, breathable fabrics (cotton). Avoid wool and rough synthetics
- Use fragrance-free, dye-free laundry detergent. Double-rinse clothes
- Keep the home cool and humid. Use a humidifier in dry months
- Wet wrap therapy for severe flares: apply moisturizer and medication, wrap the area in a damp layer of cotton, then a dry layer on top. This can dramatically reduce inflammation and itching
- For moderate to severe eczema not controlled by the above, doctors may prescribe topical calcineurin inhibitors (tacrolimus, pimecrolimus), oral antihistamines for itching, or other specialized treatments

When to See a Doctor

- Eczema is not responding to consistent moisturizing and over-the-counter treatments
- Signs of skin infection: increased redness, warmth, swelling, oozing, crusting, pus, or fever
- Eczema is affecting the child's sleep, daily activities, or quality of life
- Widespread or severe flare-up
- Eczema near the eyes

Prevention of Flare-ups

- Consistent daily moisturizing even when skin looks clear
- Identify and avoid personal triggers through an elimination approach
- Maintain a consistent skincare routine
- Manage stress and ensure adequate sleep

Source: MedlinePlus/NIH, AAP Eczema Guidelines, National Eczema Association

12. Head Lice

What Are Head Lice

Head lice are tiny parasitic insects that live on the human scalp and feed on blood. They are extremely common, especially in children ages 3 to 11. An estimated 6 to 12 million infestations occur each year in the United States. Head lice are not a sign of poor hygiene or dirty living conditions. They do not carry or spread disease. Lice cannot jump or fly. They spread almost exclusively through direct head-to-head contact and less commonly through sharing personal items like hats, combs, or pillows. Dogs, cats, and other pets do not spread human head lice.

Identification

- Lice: tiny grayish-white or tan insects about the size of a sesame seed, found on the scalp, behind the ears, and near the neckline. They move quickly and avoid light
- Nits (lice eggs): tiny oval-shaped eggs attached firmly to the hair shaft near the scalp, usually within a quarter inch. They are yellowish-white to brown and may look like dandruff, but unlike dandruff they cannot be easily brushed off
- Itching: the most common symptom, caused by an allergic reaction to lice saliva. Itching may not start for 4 to 6 weeks after the first infestation
- Small red bumps on the scalp, neck, and shoulders from scratching
- A tickling feeling of something moving in the hair

Treatment

- Over-the-counter pediculicide (lice-killing) shampoos: permethrin lotion 1 percent (Nix) is the most common first-line treatment. Apply to damp hair after washing with regular shampoo (no conditioner). Leave on for 10 minutes, then rinse. Retreat in 7 to 10 days
- Prescription treatments if OTC fails: malathion lotion, benzyl alcohol lotion, ivermectin lotion, or spinosad suspension. Follow doctor's instructions exactly
- Nit combing: after treatment, use a fine-toothed nit comb (metal combs work best) to remove nits from wet hair. Section hair and comb from scalp to ends. Wipe comb on a paper towel after each pass. Repeat every 2 to 3 days for 2 to 3 weeks
- Retreat in 7 to 10 days: most treatments do not kill all nits, so a second treatment is necessary to kill newly hatched lice
- Do not use home remedies such as mayonnaise, olive oil, petroleum jelly, or kerosene. These are not proven effective and some are dangerous
- Do not use lice treatments on children under 2 months without consulting a doctor

Household Cleaning

- Machine wash all bedding, clothing, and towels used by the infected person in the 48 hours before treatment in hot water (130 F) and dry on high heat for at least 20 minutes
- Items that cannot be washed: seal in a plastic bag for 2 weeks (lice and nits die without a human host)
- Soak combs, brushes, and hair accessories in hot water (at least 130 F) for 10 minutes
- Vacuum furniture and car seats where the child's head rested
- Do not use fumigant sprays or insecticide treatments on furniture. They are unnecessary and can be toxic

Common Myths

- Myth: Lice can jump or fly. Fact: lice can only crawl. They spread through direct head-to-head contact
- Myth: Lice prefer dirty hair. Fact: lice actually prefer clean hair because it is easier to attach to
- Myth: You need to deep clean the whole house. Fact: lice die within 1 to 2 days without a human host. Focus on items that contacted the head in the past 48 hours
- Myth: Nits on hair shafts far from the scalp are a current infestation. Fact: nits more than a quarter inch from the scalp are likely already hatched or dead

School Policies

The AAP and CDC do not recommend no-nit policies (keeping children out of school until all nits are removed). Children with active lice can finish the school day and start treatment at home that evening. They can return to school after the first treatment. The goal is to minimize disruption to the child's education while managing the infestation effectively.

Source: MedlinePlus/NIH, CDC Head Lice Information, AAP Head Lice Guidelines

13. Chickenpox (Varicella)

What Is Chickenpox

Chickenpox is a highly contagious disease caused by the varicella-zoster virus (VZV). Before the varicella vaccine became available in 1995, chickenpox was one of the most common childhood diseases, infecting nearly all children. Now, thanks to vaccination, chickenpox has become much less common. The disease causes an itchy, blister-like rash. It is usually mild in healthy children but can be serious in infants, adolescents, adults, pregnant women, and people with weakened immune systems.

Symptoms

Symptoms usually appear 10 to 21 days after exposure (average 14 to 16 days).

- The rash is the hallmark symptom and progresses through stages: flat red spots that become raised bumps (papules), then fluid-filled blisters (vesicles), then crusted scabs. All stages may be present at the same time on different parts of the body
- The rash typically starts on the chest, back, and face, then spreads to the rest of the body, including inside the mouth, eyelids, and genital area
- Intensely itchy (the defining discomfort of chickenpox)
- Fever, usually 101 to 102 F, starting 1 to 2 days before or at the same time as the rash
- Tiredness, loss of appetite, and headache
- New spots continue to appear for 3 to 5 days. The total course from first spot to all scabs is usually 5 to 10 days

Contagion

Chickenpox is extremely contagious. A person with chickenpox is contagious from 1 to 2 days before the rash appears until all blisters have crusted over (usually 5 to 7 days after the rash starts). The virus spreads through airborne respiratory droplets, direct contact with the fluid from blisters, and less commonly by touching contaminated objects. Over 90 percent of unvaccinated people who are exposed to the virus will become infected.

Treatment and Home Care

For healthy children, chickenpox is usually treated at home with comfort measures.

- Manage itching: calamine lotion applied to spots, lukewarm oatmeal baths (colloidal oatmeal bath products like Aveeno), cool compresses on itchy areas
- Oral antihistamines such as diphenhydramine (Benadryl) or cetirizine (Zyrtec) can help reduce itching, especially at night
- Keep fingernails trimmed short to prevent scratching, which can cause scarring and secondary skin infection. Consider cotton gloves at night for young children
- Acetaminophen for fever. Do NOT give ibuprofen during chickenpox as it may increase the risk of severe skin complications. NEVER give aspirin due to risk of Reye's syndrome
- Encourage fluids and soft, bland foods. If mouth sores are present, avoid salty or acidic foods
- Dress the child in loose, comfortable, soft clothing

- Antiviral medication (acyclovir) may be prescribed for teenagers, adults, people with chronic skin or lung disease, those on steroid therapy, or anyone at higher risk of complications. It is most effective when started within 24 hours of the rash appearing

Complications to Watch For

- Bacterial skin infection: if skin around a blister becomes increasingly red, warm, swollen, or tender, or if pus develops, this may indicate secondary infection requiring antibiotics
- Pneumonia (especially in adolescents and adults)
- Encephalitis (brain inflammation, rare but serious)
- Dehydration from fever and poor fluid intake

When to See a Doctor

- Fever lasts longer than 4 days or rises above 102 F
- Rash spreads to one or both eyes
- Rash areas become very red, warm, swollen, or tender (signs of secondary infection)
- Child becomes disoriented, has a stiff neck, difficulty breathing, or persistent vomiting
- Any newborn, infant, or person with a weakened immune system who is exposed to or develops chickenpox

Prevention

- Varicella vaccine: given in two doses. First dose at 12 to 15 months, second dose at 4 to 6 years. The vaccine is about 90 percent effective at preventing chickenpox and almost 100 percent effective at preventing severe chickenpox
- If an unvaccinated child is exposed to chickenpox, the vaccine given within 3 to 5 days of exposure may prevent illness or reduce severity
- Keep infected children home and away from unvaccinated individuals, infants, pregnant women, and immunocompromised persons until all blisters have crusted over

Source: MedlinePlus/NIH, CDC Varicella Information, AAP Varicella Guidelines

14. Childhood Immunization

What Are Immunizations

Immunizations (vaccines) help protect children from serious and potentially life-threatening diseases by training the immune system to recognize and fight specific germs. Vaccines contain weakened or inactive parts of a germ (or blueprints for making a protein from the germ) that trigger an immune response without causing the disease. After vaccination, if the child is later exposed to the actual disease, their immune system recognizes it and can fight it off quickly. Childhood vaccines are among the most thoroughly studied medical interventions and are recommended by the CDC, AAP, and WHO.

Recommended Vaccine Schedule (Birth to 6 Years)

The following vaccines are recommended as part of the routine childhood immunization schedule in the United States. The exact timing and number of doses vary by vaccine.

- Hepatitis B (HepB): 3 doses. First dose at birth, second at 1 month, third at 6 to 18 months
- Rotavirus (RV): 2 or 3 doses depending on brand. Starting at 2 months
- DTaP (Diphtheria, Tetanus, Pertussis): 5 doses. At 2, 4, 6, 15 to 18 months, and 4 to 6 years
- Haemophilus influenzae type b (Hib): 3 or 4 doses depending on brand. Starting at 2 months
- Pneumococcal (PCV): 4 doses. At 2, 4, 6, and 12 to 15 months
- Inactivated Poliovirus (IPV): 4 doses. At 2, 4, 6 to 18 months, and 4 to 6 years
- Influenza (Flu): annually starting at 6 months
- MMR (Measles, Mumps, Rubella): 2 doses. At 12 to 15 months and 4 to 6 years
- Varicella (Chickenpox): 2 doses. At 12 to 15 months and 4 to 6 years
- Hepatitis A (HepA): 2 doses. Starting at 12 months, with second dose 6 months later
- COVID-19: per current CDC recommendations based on age
- RSV immunization (Nirsevimab): for infants entering their first RSV season

Vaccine Schedule (7 to 18 Years)

- Tdap (Tetanus, Diphtheria, Pertussis booster): 1 dose at 11 to 12 years
- HPV (Human Papillomavirus): 2 or 3 doses starting at age 9 to 12
- Meningococcal (MenACWY): first dose at 11 to 12 years, booster at 16 years
- Meningococcal B (MenB): based on shared clinical decision-making, 16 to 23 years
- Annual influenza vaccine
- Catch-up doses for any missed vaccines from the childhood schedule

Common Side Effects

Vaccine side effects are usually mild and temporary.

- Pain, redness, or swelling at the injection site (most common)
- Low-grade fever
- Fussiness or irritability
- Fatigue or decreased appetite

- These side effects typically last 1 to 2 days and can be managed with a cool compress on the injection site and acetaminophen or ibuprofen if needed

When to Call the Doctor After Vaccination

- Fever above 105 F
- Signs of a severe allergic reaction (anaphylaxis): difficulty breathing, swelling of the face or throat, rapid heartbeat, dizziness, hives all over the body. This is extremely rare and almost always occurs within 15 to 30 minutes of vaccination
- Unusual behavior, prolonged crying (more than 3 hours), or high-pitched crying
- Seizure
- Any symptom that concerns you

Common Concerns Addressed

Vaccines do not cause autism. This claim originated from a 1998 study that was later retracted due to serious ethical violations and scientific fraud. The lead author lost his medical license. Numerous large-scale studies involving millions of children have found no link between vaccines and autism.

It is safe to give multiple vaccines at the same visit. The recommended schedule is designed so that vaccines are given at the ages when children are most vulnerable to the diseases and when their immune systems will respond most effectively. Delaying vaccines leaves children unprotected during the period when they are most at risk.

Vaccine ingredients such as thimerosal, aluminum salts, and formaldehyde are present in extremely small amounts that are safe. Thimerosal has been removed from or reduced in nearly all childhood vaccines as a precautionary measure. The amount of aluminum in vaccines is less than what a baby is exposed to through breast milk or formula. Formaldehyde is used during manufacturing and is present in trace amounts far below what the body produces naturally.

Catch-Up Vaccination

If a child has missed any vaccines, they can catch up at any age. A healthcare provider can determine which vaccines are needed based on the child's age and vaccination history. It is never too late to get back on schedule. There is no need to restart a vaccine series if doses have been missed; simply pick up where you left off.

Source: MedlinePlus/NIH, CDC Immunization Schedule 2025, AAP Immunization Guidelines, WHO Immunization Information